ORANGE UNIT 4

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SECTION:4K

Building PES-VCS — A Version Control System from Scratch

PHASE 1:

SCREENSHOT1:

```
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ./test_objetcs
bash: ./test_objetcs: No such file or directory
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

SCREENSHOT 2:

```
bash: ./test_objetcs: No such file or directory
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

PHASE2:

A PART

SCREENSHOT1:

```
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

B PART

```
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ xxd .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476 | head -20
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

PHASE3:

A PART:

```
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ make pes
make: 'pes' is up to date.
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ulimit -s unlimited
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ echo "hello" > file1.txt
echo "world" > file2.txt
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ./pes add file1.txt file2.txt
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  object.c
  untracked:  Makefile
  untracked:  commit.h
  untracked:  tree.h
  untracked:  test_objects
  untracked:  test_tree
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  pes.c
  untracked:  README.md
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  commit.c
  untracked:  test_objects.c
  untracked:  pes.h
  untracked:  .gitignore

sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

B PART:

```
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ cat .pes/index
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776322923 6 file1.txt
100664 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776322923 6 file2.txt
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

PHASE 4:

A PART:

```
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ ./pes log
commit b1b8db332b4e60bcda4a5da6d7da482debe249bd4027ba371e62e8a8c1e3120a
Author: PES User <pes@localhost>
Date:   1776324043

    Add farewell

commit f9253099d5dc27d3186d25ac7690347938f89b8eb28de357b2a4074464e111a7
Author: PES User <pes@localhost>
Date:   1776324043

    Add worldell

commit 3674377f9579aaf9dce3b0d2a71eebc2aecdcaf68084fe4fb92b9684660de5ca
Author: PES User <pes@localhost>
Date:   1776324043

    Initial committial commiQ

sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

B PART:

```
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/36/74377f9579aaf9dce3b0d2a71eebc2aecdcaf68084fe4fb92b9684660de5ca
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/b1/b8db332b4e60bcda4a5da6d7da482debe249bd4027ba371e62e8a8c1e3120a
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f9/253099d5dc27d3186d25ac7690347938f89b8eb28de357b2a4074464e111a7
.pes/refs/heads/main
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

C PART:

```
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ cat .pes/refs/heads/main
b1b8db332b4e60bcda4a5da6d7da482debe249bd4027ba371e62e8a8c1e3120a
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
sujan-anjankumar@sujan-anjankumar-IdeaPad-3-15IAU7:~/Desktop/PES1UG24CS632-pes-vcs$
```

PHASE 5 AND 6

Q5.1 — How to implement `pes checkout <branch>`

Answer:

A branch in PES is represented as a file inside `.pes/refs/heads/` containing

a commit hash. The `HEAD` file either points to a branch or directly to a commit.

To implement `pes checkout <branch>`, the following steps are required:

1. **Update HEAD**
   - Modify `.pes/HEAD` to point to the selected branch:

     ref: refs/heads/<branch>

2. **Read branch reference**
   - Open `.pes/refs/heads/<branch>` and read the commit hash.
3. **Load commit**

- Use `commit_parse` to extract the root tree hash from the commit.

4. **Reconstruct working directory**
   - Recursively read the tree objects using `object_read`
   - Create directories and files as per tree entries
   - Write blob contents to files
5. **Update index**
   - Replace current index with the tree contents of the checked-out commit

Q5.2 — Detect dirty working directory

Answer:

A dirty working directory means that the current files differ from the staged version in the index.

To detect this:

1. **Compare each tracked file (from index) with working directory**
   - Read file contents
   - Compute hash using `object_write` logic (without storing)
   - Compare with stored hash in index
2. **Check metadata differences**
   - Compare file size and modification time
3. **Conflict detection**

- If file is modified AND target branch has different version conflict

4. **Decision**
   - If any file differs refuse checkout

Q5.3 — Detached HEAD

Answer:

A detached HEAD occurs when `HEAD` contains a commit hash instead of a branch reference.

Q6.1 — Finding and deleting unreachable objects

Answer:

Over time, the object store accumulates objects not referenced by any branch.

Algorithm:

1. **Start from all branch heads**
   - Read `.pes/refs/heads/*`
2. **Traverse commits**
   - For each commit:
     - Mark commit as reachable
     - Traverse parent commits
     - Traverse associated tree
3. **Traverse tree objects**
   - For each tree:
     - Mark it
     - Traverse subtrees and blobs
4. **Mark phase**
   - Maintain a set (e.g., hash table) of reachable object hashes
5. **Sweep phase**

- Iterate over `.pes/objects/`
- Delete objects NOT in reachable set

Data structure:

- Hash set (efficient lookup O(1))

Estimation:

For:

- 100,000 commits
- 50 branches

You may visit:

- All commits in history (~100,000)
- Trees and blobs per commit

Total objects visited: **hundreds of thousands**

Q6.2 — Why GC is dangerous during commit

Answer:

Running garbage collection during a commit can lead to race conditions.

Race condition scenario:

1. Commit process starts:
   - Writes blob and tree objects
   - Has not yet updated branch reference
2. GC runs simultaneously:
   - Sees these objects as unreachable
   - Deletes them
3. Commit finishes:
   - Updates reference to objects that no longer exist

Result: repository corruption

How real Git avoids this:

1. **Atomic updates**
   - **Uses temp files and rename**
2. **Locking**
   - **Prevents GC from running during commit**
3. **Grace period**
   - **Objects are not immediately deleted**
4. **Reference safety**
   - **Objects are only removed if unreachable for a long time**